

InSAR-Assisted Random Forest Landslide Susceptibility Mapping for Joshimath, Uttarakhand

Athif sayyaf - Stage 2 Mini-Project

Abstract—This work presents a landslide susceptibility mapping (LSM) workflow for the Joshimath region of Uttarakhand using Digital Elevation Model (DEM) terrain layers, Sentinel-2 Normalized Difference Vegetation Index (NDVI), and Interferometric Synthetic Aperture Radar (InSAR) line-of-sight (LOS) velocity. A Random Forest model was trained using spatial cross-validation, because nearby pixels can make the accuracy look better than it really is if a normal random split is used. The first model used the usual landslide predictors: elevation, slope, aspect, Topographic Wetness Index (TWI), and NDVI. The second model used the same predictors with InSAR velocity added as a moving-ground layer. The model without InSAR gave an Area Under the Receiver Operating Characteristic Curve (AUROC) of 0.872 and an F1-score of 0.828. After adding InSAR velocity, AUROC increased to 0.893 and F1-score increased to 0.857. This shows a small but useful improvement when deformation information is added to the terrain and optical data.

1. INTRODUCTION

Joshimath is a mountain town in Uttarakhand, located in steep Himalayan terrain. The slopes around the town are affected by high relief, valley cutting, road construction, settlements, and rainfall. Because of these conditions, landslide susceptibility mapping (LSM) is useful for identifying areas that may behave like known landslide zones.

Most susceptibility maps are made from fairly static layers such as elevation, slope, aspect, Topographic Wetness Index (TWI), and vegetation condition. These layers explain the shape and surface condition of the terrain, but they do not show whether the ground is actually moving. This is why Interferometric Synthetic Aperture Radar (InSAR) line-of-sight (LOS) velocity was included in this study. InSAR velocity can show slow ground displacement, so it works as a dynamic layer instead of only describing the terrain. The main aim of this work is to check whether this moving-ground information improves a Random Forest landslide susceptibility model for Joshimath.

2. STUDY AREA, INVENTORY AND GIS STACK

The study area is the Joshimath region in Uttarakhand. For the analysis, a Geographic Information System (GIS) stack was prepared using elevation, slope, aspect, Topographic Wetness Index (TWI), Sentinel-2 Normalized Difference Vegetation Index (NDVI), a landslide inventory mask, and Interferometric Synthetic Aperture Radar (InSAR) line-of-sight (LOS) velocity. These layers were brought to the same Coordinate Reference System (CRS), pixel size, and spatial extent before sampling. This step is important because a small mismatch between layers can place a training point on the wrong pixel.

The landslide inventory was used as the target information for training the model. The mapped landslide locations were converted into a binary raster, where landslide pixels were marked as presence and the selected background pixels were marked as absence. Background samples were taken away from the mapped landslide pixels so that the model did not learn from areas too close to known failures. This inventory layer was then used with the predictor stack to build the training table.

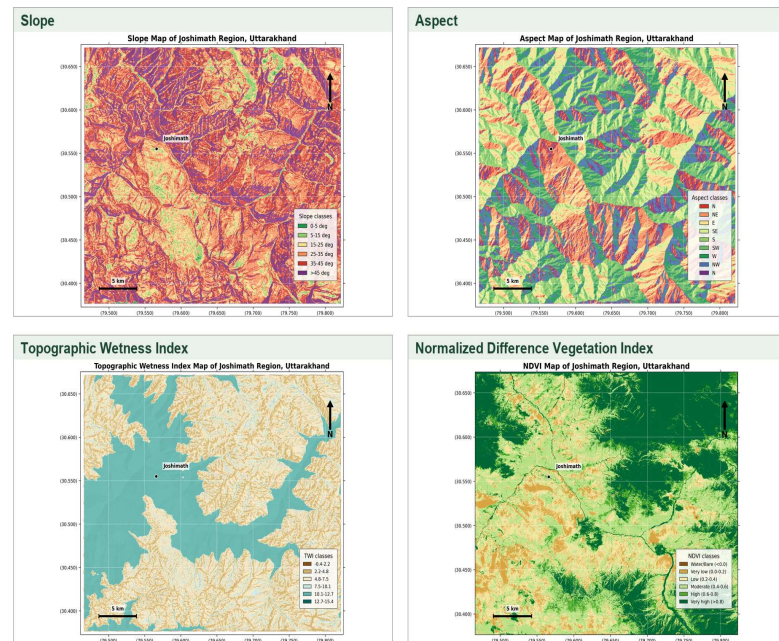


Fig. 1. GIS stack layers used for terrain and optical susceptibility modelling.

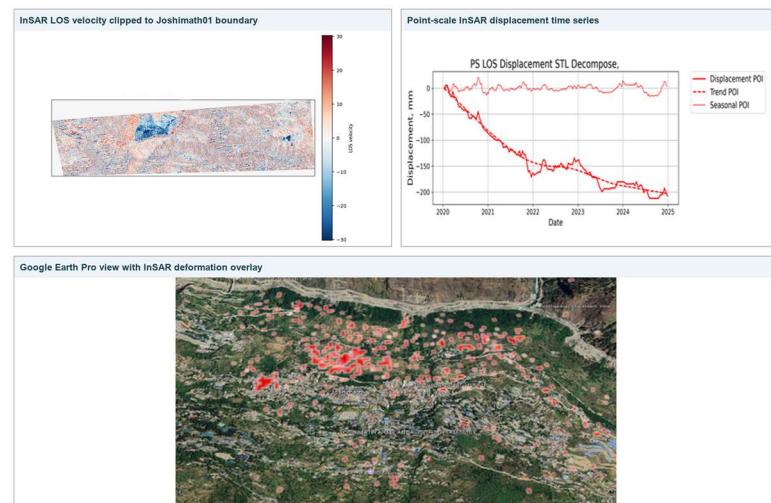


Fig. 2. InSAR LOS velocity, displacement time series and Google Earth Pro deformation overlay added after the GIS stack layer.

3. METHODOLOGY

Two Random Forest models were trained and compared. The first model used only elevation, slope, aspect, Topographic Wetness Index (TWI), and Normalized Difference Vegetation Index (NDVI). The second model used the same layers plus Interferometric Synthetic Aperture Radar (InSAR) line-of-sight (LOS) velocity. Random Forest was chosen because it is suitable for a small geospatial dataset, it can handle nonlinear relations between landslides and terrain, and it also gives feature-importance values that are easy to discuss.

For validation, spatial cross-validation was used instead of a random train-test split. The samples were grouped into spatial blocks, and each fold tested the model on blocks that were not used for training. This was done because nearby pixels can have very similar slope, elevation, vegetation, and deformation values. If nearby pixels are split randomly between training and testing, the model accuracy can become too optimistic. The final scores were calculated from the out-of-fold predictions using Area Under the Receiver Operating Characteristic Curve (AUROC), F1-score, precision, and recall.

Feature importance was taken from the trained Random Forest model. In simple terms, it shows how much each input layer helped the trees make their splits. I used this to compare the role of the normal terrain and optical layers with the added InSAR velocity layer. This helped in understanding whether the improvement came only from the map pattern or whether the deformation layer was actually contributing to the model.

4. RESULTS

Experiment	AUROC	F1	Precision	Recall	Samples
Without InSAR	0.872	0.828	0.800	0.857	28
With InSAR	0.893	0.857	0.857	0.857	28

The model without InSAR reached an AUROC of 0.872 and an F1-score of 0.828. When InSAR velocity was added, AUROC increased to 0.893 and the F1-score increased to 0.857. Precision also improved from 0.800 to 0.857, while recall stayed the same at 0.857. This means the InSAR model reduced some false positives without missing more landslide samples. Both susceptibility maps are shown using the same 0-1 scale, so the two outputs can be compared directly.

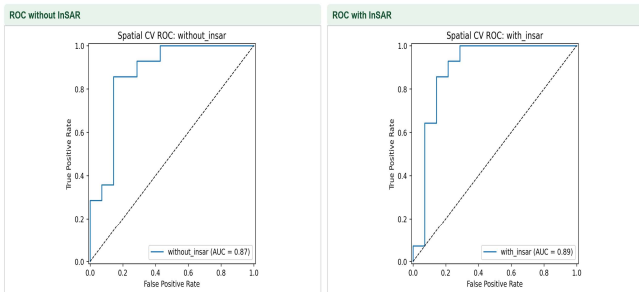


Fig. 2. ROC comparison for Random Forest models without and with InSAR.

Confusion matrix comparison

Without InSAR		With InSAR	
11 TN	3 FP	12 TN	2 FP
2 FN	12 TP	2 FN	12 TP
AUROC=0.872, F1=0.828		AUROC=0.893, F1=0.857	

Fig. 3. Confusion matrix comparison for the two experiments.

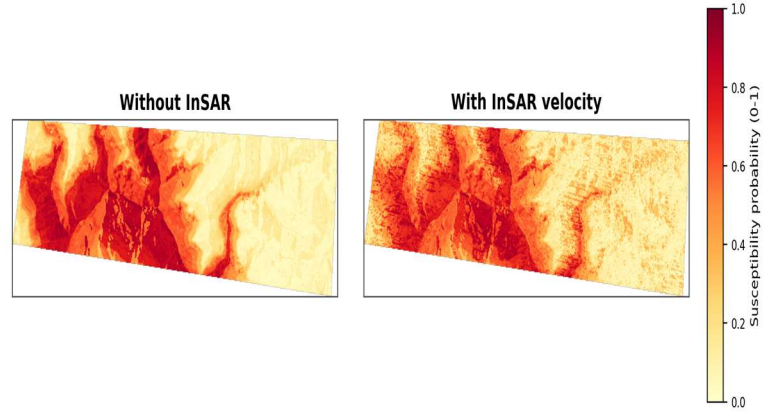


Fig. 3. Without-InSAR and with-InSAR susceptibility maps for the Joshimath01 study boundary using the same color scale.

5. DISCUSSION

The high-susceptibility areas mostly appear along steep valley-side terrain and around slopes close to the settlement zone. This makes sense for Joshimath because many unstable areas are expected on steep, cut, or disturbed slopes. Elevation was still the strongest static predictor, but InSAR LOS velocity added useful information about actual ground movement. This is important because two slopes may look similar in the terrain layers, but one of them may also be slowly deforming.

There are also some clear limitations. The number of samples inside the common InSAR coverage was small, so the result should be treated as a pilot study rather than a final operational map. A proper coherence layer was not available for masking unreliable InSAR pixels. Also, the non-landslide samples were assumed to be stable background samples, not field-checked stable points. With more time, I would add a larger landslide inventory, a mean coherence threshold, lithology, distance to roads and streams, rainfall, and an independent validation area.

6. CONCLUSION

This study shows that adding InSAR LOS velocity can improve landslide susceptibility mapping for the Joshimath area, even if the improvement is moderate. The result supports the idea that static terrain layers and dynamic ground-motion information should be used together where InSAR data are available. At the same time,

the result should be interpreted carefully because the sample size and InSAR coverage are limited.

REFERENCES

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